

Nachtalb: A multisensory Neurofeedback VR-Interface

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Figure 1: Visual Translation of the Frequency Range

ABSTRACT

Nachtalb is an immersive interface that enables brain-to-brain interaction using multisensory feedback. With the help of the g.tec Unicorn Hybrid Black brain-computer-interface (BCI), brain-activity-data is measured and translated visually with the Oculus Quest 2, tactilely with the bHaptics TactSuit and auditorily with 3D Sound. This intends to create a feedback loop that turns brain activity from data-input into sensory output which directly influences the brain activity data-input again.

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1 BACKGROUND

Inside the brain, neurons interact in form of electrochemical signals that generate electromagnetic fields. Based on the frequency and amplitude range, these electric fields are classified into five distinct band-limited signal patterns known as brainwaves: delta, theta, alpha, beta and gamma [Nunez and Srinivasan 2006].

These neural oscillations are the foundations of our perception. Yet, these processes themselves are not directly perceptible to us. We can only see, hear, feel and recognize the effects of emotional or cognitive, motoric or physical dimensions.

In fact, mood, emotion and feelings correlate with the brainwaves and thus inferences can be drawn from brain activity to mental states [Liu et al. 2010]. Non-invasive rhythmic stimulation can also affect the frequency of brainwaves, theoretically defined as

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brainwave entrainment. It can be observed that auditory, visual or tactile rhythmic stimulation at a certain frequency correlates with the parallel measured frequency of the brainwaves [Argento et al. 2017].

2 RELATED WORK

Traditionally, BCI technology is used in a clinical context. In neurofeedback therapy, brain activity is presented to subjects as feedback, which the subjects use to learn and improve their mental strategy [Sitaram and Ros 2017]. Yet, BCI technology has found application in an artistic and design context as well. Sobell introduced "Interactive Brain Wave Drawing" already in 1973, plotting EEG-Data of two people superimposed on their faces [Sobell 2002]. Sobell was able to visualize the nonverbal influence of two people on each others brainwaves. More recently, De Smedt and Meschaert developed a system called "VALENCE", which monitors the participant's alpha-waves (indicator of relaxation) and valence (indicator of emotion and arousal) as animated particles [Smedt and Menschaert 2012]. Parenthoen et al. as well as Fernandez et al. combined BCI and Virtual Reality feedback in their work. While "The Sea Is Your Mirror" is an immersive installation that translates brainwaves into an animated wave [Parenthoen et al. 2015], "Deep Reality" also monitors the participant's heart rate and provides subtle feedback to influence their level of relaxation [Fernandez et al. 2019]. Applications with a broader gaming character are "Alphawow", where the avatar's character changes its behaviour according to the player's state of relaxation [van de Laar et al. 2013] and "Paint with Your Mind", an interactive installation that enables visitors of a museum to control the generation of lines, colors, and movements of characters in traditional Chinese paintings using real-time EEG-Data [Chen et al. 2021]. Still, these applications usually utilize visual feedback, but the requirement for utilizing a person's visual system is simply that this system is still functional [Breitwieser et al. 2012]. Technical advances have made it possible to implement auditory, tactile or multisensory feedback as well. Recent work, therefore, explores other sensory modalities also in VR environments and hypothesizes that multisensory stimulation may improve the quality of performance [Triantafyllidis et al. 2020]. It has also been suggested that multisensory stimuli are perceived more precisely and

faster than unisensory stimuli [Sigrist et al. 2013]. Still, it remains an open question to which extent it is justified to utilize multisensory feedback systematically and therefore further investigations are needed, that clear the question for interaction designers, in which way multisensory stimuli can or should be implemented in future interfaces that utilize BCI and VR technology.

3 SYSTEM OVERVIEW

While non-invasive BCIs reveal our neurological processes, they do not actively interfere with our perception. VR, on the other hand, allows us to experience natural-like stimuli and thus can externally influence our perception on immersive dimension.

Based on this relation, we designed an interface that implements real-time BCI data processing and multisensory VR feedback. The user's measured brain activity data is translated into visual, auditory and tactile rhythms to finally act as direct stimuli in return.

3.1 User Experience and Visuals

For the user, the interface appears as follows: First, the rendered sphere is viewed from outside. The sphere consists of a swarm of particles pulsating at the user's own main brain frequency. Additionally, an impulse sound is perceptible, which is synchronous to this frequency. If the user moves into the sphere, the impulse sound becomes clear, furthermore the TactSuit starts to emit a vibration impulse, which is also synchronized with the main brain frequency. The active decision to engage with one's own brain activity is rewarded with the overall experience: immersion inside the translation of one's neurological processes, to which, in turn, the inputs to tactile, auditory and visual senses are coupled.

3.2 Technical Realization

The raw EEG data measured in real-time by the Unicorn Hybrid Black is sent to Simulink (MATLAB R2020a) via Bluetooth. The raw data from each electrode is merged and first filtered through a 50 Hz and a 60 Hz notch filter to eliminate power line interference. Five bandpass filters divide the entire spectrum into five frequency bands: delta, theta, alpha, beta and gamma, in order to finally determine the average activity within the respective frequency bands. In the next step, this results in the frequency band with the highest average activity. In the last step, the frequency with the highest activity (main-frequency) from this frequency band is defined by a power spectral density analysis (based on Welch's method) and streamed in real-time to Unity through the Lab Streaming Layer.

Visual feedback is provided by the particle system, made with Unity. The opacity of the particles is pulsating in the main-frequency. Reacting to changes of the main-frequency, the movement and appearance of the sphere relatively changes. In Figure 1, the appearance at 5 Hz (A), 10 Hz (B), 20 Hz (C) and 40 Hz (D) is illustrated.

For auditory feedback, FMOD is implemented in Unity. The audio-frequency of the sound is modulated by a Low-Frequency-Oscillator (LFO) equivalent to the main-frequency. This generates the pulsing effect on an auditory level. While the head of the user is outside the sphere, a Low-Pass-Filter (LPF) damps the sound and only when the user is located inside the sphere, the full sound is perceivable.

Tactile feedback is provided by the bHaptics TactSuit, while the user is located inside the sphere. The integrated Software bHaptic Player is supporting the audio-based tactile feedback, linking it to the auditory-feedback. While the LPF is still active, no tactile feedback is provided. The LFO indirectly modulates the vibration rhythm, turning the vibration on and off in the main-frequency.

4 CONCLUSION AND FUTURE WORK

We present the interaction of BCI and multisensory VR technology with the capability of offering a new form of encounter with yourself. Following the concept of the feedback loop, the source of the user's perception consists in this case of the translation of their neural oscillations. This means, that combining these technologies enables the user's brain to interact with itself on an more and more immersive dimension.

Nachtalb provides answers regarding its technical and aesthetic realization while focusing on the experience itself. It much more aims to raise further discussions on the utilization of BCIs and multisensory VR as both will continue to improve. Future investigations are needed to clear which potential lies in connecting these technologies.

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